

LDRM — Liability Determination Rationale Module

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Architecture: LIGA™ — Liability Governance Architecture
Parent Standard: Liability Determination Standard (LDS)
Category: Governance & Enforcement
Subcategory: Liability Governance
Type: Core Module
Governed Space: Liability Determination Rationale
Module ID: LIGA-LDS-LDRM-001
Origin ID: OOF.CM.LIGA.07.04.01
Version: 1.0
Status: Canonical · Open Standard
Effective Date: 23 September 2026
Compatibility: OOF® Methodology OS™ · Modular OOF® Architecture Ecosystem
AI-Readable: Yes
Authority: OOF® — OriginOpen® Foundation
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition

The Liability Determination Rationale Module (LDRM) governs how the reasoning underlying a Liability Attachment Determination is explicitly structured, associated, documented, bounded, traceable, and preserved.

LDRM establishes the governed rationale explaining why a specific Liability Attachment State follows from the applicable Liability Basis, Conditions, Subjects, Relationships, Connections, Determination Inputs, and Determination Evaluation.

It ensures that a Liability Determination does not exist merely as an unexplained outcome.

LDRM does not create or change the Attachment State merely by documenting it.

It governs the transition:

Governed Liability Attachment State → Rationale Formation → Basis & Evaluation Association → Supporting & Contradicting Factor Mapping → Uncertainty & Limitation Representation → Rationale Sufficiency → Governed

Liability Determination Rationale

Its foundational distinction is:

Liability Determination Outcome ≠ Liability Determination Rationale.

Operational Role

The operational role of LDRM is:

to make the reasoning behind a Liability Determination explicit enough that the Determination can be understood, reviewed, reconstructed, challenged, compared, corrected, and governed through time.

LDAM determines the Attachment State.

LDRM explains the governed basis for that state.

LDRM does not independently re-perform the complete Determination Evaluation.

Position in the Parent Standard Flow

The internal LDS progression is:

Liability Determination Input → Liability Determination Evaluation → Liability Determination & Attachment → Liability Determination Rationale → Liability Determination Versioning LDRM governs:

Liability Determination Rationale

Its principal input is:

Governed Liability Attachment State

Its principal output is:

Governed Liability Determination Rationale

Its canonical handoff is:

Governed Liability Determination Rationale → Liability Determination Versioning

Central Operational Question

Why does the established Liability Attachment State follow from the governed Liability structure, and can that reasoning be explicitly reconstructed and reviewed?

Rationale Requirement

Every material Liability Attachment Determination should possess a sufficiently explicit rationale.

A Determination should not depend solely on:

an outcome label, an unexplained conclusion, institutional status, human intuition, automated output, or narrative assertion.

Therefore:

Determination State without Rationale ≠ Fully Governed Determination.

Rationale Object

Each rationale should be associated with a specific Liability Determination.

The association should preserve:

Determination identity, Attachment identity, Subject, Relationship, Liability Object, Liability Basis, and Determination Scope.

Therefore:

General Explanation ≠ Determination-Specific Rationale.

Rationale Scope

The rationale should explain only the governed Determination to which it applies.

It should not silently extend a Subject-specific or Basis-specific determination to other parts of the Liability Network.

Therefore:

Rationale Scope ≠ Complete Liability Network Scope by default.

Basis Association

The rationale should identify the Governed Liability Basis under which the Attachment Determination was made.

This allows the reasoning to answer:

Why could Liability potentially exist under this Basis, and why did the evaluated structure produce this Attachment State?

Condition Association

The rationale should identify materially relevant Liability Conditions and their states.

Where a Condition:

was satisfied, failed, was modified, was subject to an Exception, remained uncertain, or remained disputed, its significance to the Determination should be representable.

Subject and Relationship Association

The rationale should identify the Liability Subject and Liability Relationship to which it applies.

This prevents reasoning concerning one Subject or Relationship from being transferred automatically to another.

Therefore:

Rationale for Subject A $\neq$ Rationale for Subject B.

Object Association

The rationale should remain associated with the Governed Liability Object relevant to the Determination.

Therefore:

Rationale concerning Object A $\neq$ Rationale concerning Object B.

Connection Association

The rationale should identify the materially relevant Governed Liability Connection Structure relied upon by the Determination.

This may include:

Connection Relationships, Chains, qualification states, and materially relevant uncertainties.

Evaluation Association

LDRM should preserve the relationship between the rationale and the Governed Liability Determination Evaluation.

The rationale should explain how the Evaluation informed the Attachment State.

Therefore:

Rationale $\neq$ Replacement for Evaluation.

Input Association

Material Determination Inputs relied upon in the rationale should remain traceable.

The rationale need not reproduce the entire input structure.

It should identify the inputs materially necessary to understand the Determination.

Supporting Factors

LDRM should identify the materially relevant factors supporting the Attachment Determination.

Supporting factors may concern:

Basis, Conditions, Relationships, Connections, operational states, Harm states, evidence-associated states, or other governed inputs.

A supporting factor does not independently become Liability.

Contradicting Factors

Material factors contradicting or limiting the Determination should also remain visible.

A governed rationale should not present only favorable information while suppressing materially contrary inputs.

Therefore:

Rationale $\neq$ Advocacy Narrative.

Limiting Factors

The rationale should identify factors that limit:

scope, confidence, applicability, interpretation, or strength of the Determination where materially relevant.

Unknown Factors

Material unknowns affecting the Determination should remain explicit.

Therefore:

Unknown ≠ Absent.

An unexplained information gap should not silently become support for either Attachment or non-Attachment.

Uncertainty

Material uncertainty should be represented within the rationale.

The rationale should identify:

what is uncertain, why it is uncertain, and how that uncertainty affects the Determination.

Therefore:

Explicit Rationale ≠ Certainty.

Disputed Elements

Where materially relevant inputs or evaluations remain disputed, the rationale should preserve that dispute.

Therefore:

Documented Rationale ≠ Resolved Dispute.

Alternative Explanations

Where materially plausible alternative evaluation paths exist, the rationale should identify them where necessary to understand the Determination.

An alternative should not be erased merely because it was not adopted.

Rejected Alternative

Where one materially relevant alternative was rejected, the rationale should preserve the governed reason for rejection where necessary.

Therefore:

Rejected Alternative ≠ Nonexistent Alternative.

Rationale Structure

A governed rationale should be capable of showing the reasoning chain:

Liability Basis → Relevant Conditions → Relevant Subjects & Relationships → Liability Connection → Material Determination Inputs → Determination Evaluation → Attachment State The rationale should make material transitions understandable.

Reasoning Link

Each material reasoning link should connect a governed input or evaluated state to the conclusion it supports.

Therefore:

Sequence of Statements ≠ Governed Reasoning Chain by default.

Rationale Traceability

The rationale should allow a qualified reviewer to trace material reasoning backward to the relevant governed objects.

Traceability should extend, where necessary, through:

Attachment State, Evaluation, Determination Inputs, Connection, Conditions, Relationships, Basis, Object, and Context.

Rationale Reconstruction

A qualified reviewer should be able to reconstruct:

what was determined, what governed structure was relied upon, which material factors supported the result, which factors contradicted or limited it, what remained unresolved, and why the Attachment State resulted.

Rationale Explainability

The rationale should be understandable without requiring access to hidden or undocumented reasoning.

Therefore:

Unrecorded Reasoning $\neq$ Governed Rationale.

Human-Readable Rationale

Where the Determination affects human decision-making or review, the rationale should be representable in a form understandable to the relevant human audience.

Human-readable explanation should preserve methodological accuracy.

Machine-Readable Rationale

Where implemented digitally, the rationale should also be capable of structured machine-readable representation.

This may preserve relationships among:

inputs, criteria, evaluations, states, and conclusions.

Human and Machine Representation

Human-readable and machine-readable representations should refer to the same governed rationale object where they represent the same Determination.

Therefore:

Different Representation $\neq$ Different Rationale by default.

Rationale Granularity

The rationale should contain enough detail to explain the Determination without reproducing every upstream governance record.

Therefore:

Rationale Completeness $\neq$ Repetition of Entire Liability Record.

Rationale Sufficiency

A rationale is sufficiently governed when it explains:

what was determined, for whom or what, under which Basis, within what scope, using which material Conditions and Connections, what Evaluation supported the result, what material supporting and contradicting factors existed, what uncertainty remained, and why the Attachment State followed.

Rationale Completeness

Rationale Completeness concerns whether all materially necessary reasoning elements are represented.

It does not require all available information to be repeated.

Therefore:

Rationale Completeness $\neq$ Information Exhaustiveness.

Rationale Consistency

The rationale should be consistent with the governed Attachment State and underlying Evaluation.

Where inconsistency exists, it should trigger correction or re-entry.

Therefore:

Rationale Contradiction $\neq$ Valid Explanation.

Outcome-Rationale Consistency

An Attached state should not possess a rationale that materially supports Not Attached without the conflict being explicitly governed.

Likewise, a Not Attached state should not conceal materially contrary Evaluation findings.

No Post-Hoc Rationale

LDRM should prevent a conclusion from being made first and an artificial rationale constructed afterward merely to justify it.

Therefore:

Post-Hoc Justification $\neq$ Governed Determination Rationale.

No Authority-Only Rationale

The fact that an authority produced a Determination does not itself explain why the Determination was made.

Therefore:

Authority $\neq$ Rationale.

Authority and rationale are distinct governance objects.

No Evidence-Only Rationale

A list of evidence does not explain how that evidence relates to the Liability Determination.

Therefore:

Evidence List $\neq$ Liability Determination Rationale.

No Outcome-Only Rationale

Statements such as:

“Liability exists” or:

“Liability does not exist” are outcomes, not rationales.

Therefore:

Conclusion $\neq$ Explanation.

No Harm-Only Rationale

The existence or severity of Harm is not a complete Liability rationale.

Therefore:

Harm Description $\neq$ Liability Determination Rationale.

No Causation-Only Rationale

An operational causal chain may be relevant but does not by itself explain the complete Liability Determination.

Therefore:

Causation Explanation $\neq$ Complete Liability Rationale.

No Responsibility-Only Rationale

Responsibility may contribute to the rationale without becoming the rationale itself.

Therefore:

Responsibility Finding $\neq$ Liability Determination Rationale.

No Claim-Only Rationale

The existence or content of a Claim does not provide a complete Liability rationale.

Therefore:

Claim Assertion ≠ Liability Determination Rationale.

Rationale for Non-Attachment

A Not Attached state requires a governed rationale just as an Attached state does.

The rationale should identify why the evaluated structure does not establish Attachment within the defined scope.

Therefore:

Non-Attachment ≠ Absence of Rationale Requirement.

Rationale for Pending State

A Pending state should identify the material dependency preventing completion.

The rationale should make clear what remains unresolved.

Rationale for Unknown State

An Unknown state should identify why the governed structure cannot establish the Attachment question.

Therefore:

Unknown State ≠ Unexplained State.

Rationale for Uncertain State

An Uncertain state should identify the material source and effect of uncertainty.

Rationale for Disputed State

A Disputed state should preserve the competing governed positions and the unresolved point of disagreement.

Therefore:

Dispute Documentation ≠ Dispute Resolution.

Multi-Subject Rationale

Where multiple Subjects receive separate Attachment States, each materially distinct Determination should possess its own rationale or an explicitly partitioned shared rationale.

Therefore:

Shared Incident ≠ Shared Liability Rationale.

Multi-Relationship Rationale

Where one Subject participates in several Liability Relationships, the rationale should preserve Relationship-specific reasoning.

Therefore:

One Subject ≠ One Universal Liability Rationale.

Multi-Basis Rationale

Where Liability is evaluated under several Bases, rationale should remain Basis-specific where the reasoning materially differs.

Therefore:

Rationale under Basis A ≠ Rationale under Basis B by default.

Liability Network Rationale

A complex Liability Network may contain several connected but distinct rationales.

A local rationale should not be represented as explaining the complete network unless it genuinely does so.

Therefore:

Local Determination Rationale ≠ Complete Liability Network Rationale.

Rationale Authority

The rationale should preserve the identity of the process, system, institution, or authority responsible for the Determination where applicable.

This allows the reasoning to remain associated with its actual governance origin.

Automated Rationale

Where an AI or automated system generates or assists with a rationale, the system should preserve sufficient traceability to:

the relevant Determination, input structure, Evaluation, methodology, system or model version where applicable, and authority structure.

Therefore:

Generated Explanation ≠ Governed Rationale by default.

Human Rationale

A human-authored explanation should still satisfy the same governance requirements for:

scope, traceability, consistency, materiality, and historical integrity.

Therefore:

Human Explanation ≠ Governed Rationale by default.

Rationale Change

A rationale may change where:

the Determination changes, the Evaluation changes, an error is corrected, new information becomes relevant, or an authority issues a revised explanation.

Changes should remain traceable.

Rationale Correction

A rationale may be corrected where its earlier representation was inaccurate or incomplete.

The correction should preserve:

the earlier rationale, corrected rationale, reason, time, and provenance.

Therefore:

Rationale Correction ≠ Historical Erasure.

Rationale Reassessment

A rationale may require reassessment after upstream governance changes.

Reassessment does not necessarily mean the earlier rationale was invalid when created.

Therefore:

Rationale Reassessment ≠ Rationale Correction.

Rationale Versioning

Material rationale changes should be versioned.

Therefore:

New Rationale Version ≠ New Liability Origin by default.

Full Determination version governance belongs to LDVM.

Rationale Re-Entry

Where rationale formation reveals a substantive problem with the Attachment Determination, Evaluation, or Determination Inputs, LDRM should trigger re-entry rather than conceal the inconsistency.

Possible routes include:

Attachment defect → LDAM Evaluation defect → LDEM Input defect → LDIM or the earliest materially affected upstream LIGA™ governance space.

Historical Rationale Integrity

Earlier rationales should remain reconstructable where they materially informed earlier Liability states.

Therefore:

Current Rationale ≠ Complete Rationale History.

Rationale Record

A governed rationale record should preserve, where applicable:

Rationale identity, Determination identity, Attachment State, Subject, Relationship, Object, Basis, scope, material Conditions, material Connections, Evaluation reference, supporting factors, contradicting factors, limitations, uncertainty, alternatives, authority, time, and version.

Governed Liability Determination Rationale

The principal output of LDRM is:

Governed Liability Determination Rationale

It represents the explicit methodological explanation connecting the governed Liability structure to the established Attachment State.

It does not replace the Attachment Determination.

Therefore:

Governed Liability Determination Rationale ≠ Liability Attachment State.

Handoff to Liability Determination Versioning

LDRM hands the:

Governed Liability Determination Rationale

together with its associated Attachment State to:

LDVM — Liability Determination Versioning Module

The next governance question becomes:

How must the complete Liability Determination remain identifiable, versioned, historically coherent, and reconstructable as it changes through time?

Minimum Implementation Framework

A conforming implementation of LDRM must provide exactly the following eight capabilities:

Rationale Identity, Object & Scope Governance

Capability to associate each rationale with the correct Determination, Attachment State, Subject, Relationship, Object, Basis, and governed scope.

Basis, Condition, Relationship & Connection Rationale Governance

Capability to explain how the materially relevant Liability Basis, Conditions, Relationships, and Connection Structure contributed to the Determination.

Evaluation & Material Input Association

Capability to trace the rationale to the Governed Liability Determination Evaluation and the material inputs necessary to understand the resulting Attachment State.

Supporting, Contradicting, Limiting & Alternative Factor Governance

Capability to preserve material support, contradiction, limitations, competing explanations, and rejected alternatives without converting the rationale into advocacy.

Uncertainty, Unknown & Dispute Rationale Governance

Capability to explain material uncertainty, unknown states, pending dependencies, and disputed positions and their effect on the Determination.

Rationale Traceability, Explainability & Sufficiency Governance

Capability to ensure that the rationale is reconstructable, understandable, sufficiently complete, and consistent with the governed Determination without requiring information exhaustiveness.

Rationale Change, Correction, Versioning & Re-Entry

Capability to preserve rationale changes, corrections, reassessments, versions, historical states, and targeted re-entry without historical erasure.

Determination Versioning Handoff

Capability to produce a Governed Liability Determination Rationale sufficiently explicit and traceable for integration into Liability Determination Versioning.

Governance Logic

The canonical LDRM governance logic is:

Governed Liability Attachment State

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Identify Determination & Rationale Scope

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Associate Liability Basis

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Associate Material Conditions

↓

Associate Subject & Relationship

↓

Associate Governed Liability Connection Structure

↓

Associate Determination Evaluation

↓

Identify Material Supporting Factors

↓

Identify Material Contradicting & Limiting Factors

↓

Preserve Unknowns / Uncertainty / Disputes

↓

Represent Material Alternatives

↓

Construct Reasoning Chain

↓

Verify Outcome-Rationale Consistency

↓

Assess Rationale Traceability

↓

Assess Rationale Sufficiency

↓

Preserve Rationale History

↓

Governed Liability Determination Rationale

↓

Handoff to Liability Determination Versioning

Governance Outputs

LDRM may produce:

Liability Determination Rationale Identity

Rationale Scope

Basis Rationale Association

Condition Rationale Association

Subject Rationale Association

Relationship Rationale Association

Connection Rationale Association

Evaluation Rationale Association

Material Input Association

Supporting Factor Record

Contradicting Factor Record

Limiting Factor Record

Unknown Factor Record

Uncertainty Rationale

Dispute Rationale

Alternative Explanation Record

Rejected Alternative Record

Reasoning Chain

Rationale Trace Record

Rationale Sufficiency State

Rationale Correction Record

Rationale Reassessment Record

Rationale Version Record

Historical Rationale Record

The principal output is:

Governed Liability Determination Rationale

Critical Distinctions

LDRM preserves the following distinctions:

Liability Determination Outcome ≠ Liability Determination Rationale Determination State without Rationale ≠ Fully Governed Determination General Explanation ≠ Determination-Specific Rationale Rationale Scope ≠ Complete Liability Network Scope by default Rationale for Subject A ≠ Rationale for Subject B Rationale concerning Object A ≠ Rationale concerning Object B Rationale ≠ Replacement for Evaluation Rationale ≠ Advocacy Narrative Unknown ≠ Absent Explicit Rationale ≠ Certainty Documented Rationale ≠ Resolved Dispute Rejected Alternative ≠ Nonexistent Alternative Sequence of Statements ≠ Governed Reasoning Chain by default Unrecorded Reasoning ≠ Governed Rationale Different Representation ≠ Different Rationale by default Rationale Completeness ≠ Repetition of Entire Liability Record Rationale Completeness ≠ Information Exhaustiveness Rationale Contradiction ≠ Valid Explanation Post-Hoc Justification ≠ Governed Determination Rationale Authority ≠ Rationale Evidence List ≠ Liability Determination Rationale Conclusion ≠ Explanation Harm Description ≠ Liability Determination Rationale Causation Explanation ≠ Complete Liability Rationale Responsibility Finding ≠ Liability Determination Rationale Claim Assertion ≠ Liability Determination Rationale Non-Attachment ≠ Absence of Rationale Requirement Unknown State ≠ Unexplained State Dispute Documentation ≠ Dispute Resolution Shared Incident ≠ Shared Liability Rationale One Subject ≠ One Universal Liability Rationale Rationale under Basis A ≠ Rationale under Basis B by default Local Determination Rationale ≠ Complete Liability Network Rationale Generated Explanation ≠ Governed Rationale by default Human Explanation ≠ Governed Rationale by default Rationale Correction ≠ Historical Erasure Rationale Reassessment ≠ Rationale Correction New Rationale Version ≠ New Liability Origin by default Current Rationale ≠ Complete Rationale History Governed Liability Determination Rationale ≠ Liability Attachment State

Cross-Architecture Boundary

LDRM may reference governed objects originating from other OOF® architectures where those objects materially contributed to Liability Determination.

For example:

ORA™ may provide Operational Reality.

HAGA™ may provide governed Harm states.

AGA™ may provide Responsibility or Accountability states.

GOA™ may provide Authority structures.

ADIT® may provide evidence and auditability structures.

VALIDOS™ may provide validation states.

OBIDENTITY® may provide identity, ownership, authority, permission, provenance, transfer, or continuity states.

AIG® and ASGA™ may provide AI or autonomous-system governance states.

PGA™ may provide Prediction states.

SIMULOS® may provide Simulation outputs.

LDRM does not reinterpret those objects inside their original governance domains.

It explains only their governed relevance to the Liability Determination.

The canonical interoperability principle is:

A governed external state may contribute to a Liability rationale without becoming Liability and without transferring governance ownership of its original domain to LIGA™.

Canonical Boundary

LDRM begins with a:

Governed Liability Attachment State

and governs the explicit rationale supporting that Determination.

LDRM governs:

Rationale Identity, Rationale Scope, Basis Association, Condition Association, Subject and Relationship Association, Object Association, Connection Association, Evaluation Association, Material Input Association, Supporting Factors, Contradicting Factors, Limiting Factors, Unknowns, Uncertainty, Disputes, Alternatives, Reasoning Structure, Traceability, Explainability, Sufficiency, Consistency, Change, Correction, Reassessment, Rationale Versioning, Historical Integrity, and Determination Versioning Handoff.

LDRM does not govern:

formation of Determination Inputs, substantive Determination Evaluation, formal Liability Attachment, Liability Scope, Liability Allocation, ongoing Liability State, Transfer, Recourse, Resolution, Discharge, or Finality.

LDRM ends when a:

Governed Liability Determination Rationale

is sufficiently explicit and traceable for Liability Determination Versioning.

Use Case 1 — Autonomous System Determination Rationale

LDAM establishes an Attached state for a Liability Relationship involving an operator and an autonomous industrial system.

LDRM links that Determination to:

the applicable Liability Basis, relevant Conditions, the operator Relationship, the qualified Connection structure, the governed Determination Evaluation, and the material operational and evidentiary inputs.

The rationale also records a remaining uncertainty concerning one autonomous-system control state and explains why that uncertainty does not change the established Attachment State under the applicable methodology.

The result is a reconstructable explanation of why the Attachment State was established rather than merely a statement that Liability exists.

Use Case 2 — Multi-Party Liability Rationale

An industrial incident produces different Attachment States for an operator, maintenance provider, component producer, and owner.

LDRM preserves separate rationales for the materially distinct Determinations.

The operator rationale may rely on one Basis and Connection structure.

The maintenance-provider rationale may rely on a different Relationship and Condition set.

The producer Determination may remain Pending because a material input remains unresolved.

The owner may be Not Attached under the evaluated Basis.

LDRM prevents the shared incident from being represented through one generic explanation and preserves the reasoning specific to each Liability Determination.

Canonical Closing Statement

A Liability Determination should never exist merely as an outcome.

Its governed reasoning must remain identifiable, bounded, traceable, reviewable, and reconstructable.

LDRM preserves the path from the applicable Liability Basis and governed inputs through Evaluation to the resulting Attachment State while keeping supporting factors, contradictions, uncertainty, limitations, and alternatives visible.

It does not create Liability.

It makes the reasoning behind Liability Determination governable.

A Determination states the outcome. A Rationale makes the outcome reconstructable.

If Liability cannot be explained through its governed structure, the Determination is not yet fully governed.

Governed Links

Liability Determination Standard (LDS)

LIGA™ Architecture Map

LDVM — Liability Determination Versioning Module

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